

Week 5 Assignment Reflection

Intelligent Anomaly Detection and AI-Generated Root Cause Analysis

During this assignment, I developed a complete anomaly detection and root cause analysis (RCA) workflow using machine learning, structured log analysis, OpenTelemetry instrumentation, and an agent-based analysis process. Instead of relying only on anomaly detection, I implemented a workflow that first identifies abnormal metrics, groups related alerts into incidents, and then generates a structured RCA report using information collected from both metrics and application logs.

One important design decision was selecting the contamination parameter for the Isolation Forest model. I evaluated several values (0.01, 0.02, and 0.10) instead of accepting the default configuration. After comparing precision, recall, and F1 score, I selected a contamination value of 0.02 because it provided the best balance between detecting true anomalies and minimizing false positives. This demonstrated that model parameters significantly influence anomaly detection quality and should be evaluated rather than chosen arbitrarily.

Another important part of the project was alert grouping. Rather than analyzing hundreds of alerts individually, I grouped related anomalies into incidents based on similar error characteristics. This reduced alert noise and made it easier for the RCA agent to analyze a single incident instead of processing many isolated alerts. The agent then used two separate tools—a metrics analysis tool and a log analysis tool—to gather evidence before generating the final root cause analysis report.

One challenge I encountered was exporting OpenTelemetry spans to a JSON file. Initially, the output file was created but remained empty because the spans had not yet been flushed before the program exited. I resolved this by updating the telemetry implementation to use a synchronous span processor and explicitly flushing the exporter after execution. This experience reinforced the importance of validating telemetry output instead of assuming instrumentation is working correctly.

Overall, this assignment demonstrated how machine learning, observability, and agentic AI can work together to improve operational troubleshooting. Instead of producing only anomaly predictions, the complete workflow correlated alerts, gathered supporting evidence, generated a structured RCA report, and recorded telemetry for the agent's reasoning process. This project strengthened my understanding of AI-assisted operations, OpenTelemetry instrumentation, and practical approaches to automated incident investigation.